

Sentiment-Adjusted Heston Monte Carlo Option Pricing Framework

Joseph Hadchiti

February 2026

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1 Introduction

This report describes a quantitative option pricing framework that combines three main components:

1. The Heston Stochastic Volatility Model
2. Monte Carlo Simulation
3. News-sentiment Adjustment (to the model’s long-run variance parameter)

Standard constant-volatility models (e.g. Black–Scholes) fail to capture the volatility smile observed in listed equity options. The Heston model addresses this by allowing instantaneous variance to itself follow a stochastic diffusion. The sentiment layer serves as an alternative data signal which introduces a forward-looking correction to the model’s long-run variance based on the average polarity of recent news articles.

This entire pipeline is self-contained and implemented in Python, performing the following end-to-end steps:

1. Fetch historical stock prices and the option chain from Yahoo Finance.
2. Retrieve recent news headlines and compute a daily sentiment score.
3. Estimate the initial variance v_0 from realised returns.
4. Adjust the Heston long-run variance θ using the sentiment score.
5. Calibrate Heston parameters $(\kappa, \theta, \sigma_v, \rho, v_0)$ jointly to market call prices via differential evolution, with the Feller condition enforced by construction through a parameter reparameterisation.
6. Run a large-scale Monte Carlo simulation of the calibrated model.
7. Price the full option chain and assess put-call parity.
8. Visualise simulated paths, variance dynamics, and model-vs-market prices.

2 The Heston Stochastic Volatility Model

2.1 Stochastic Differential Equations

Under the risk-neutral measure $\mathbb{Q}$, the Heston model specifies the joint dynamics of the asset price S_t and its instantaneous variance v_t as:

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^{(2)}, \quad (2)$$

where:

- r is the continuously compounded risk-free rate,
- $\kappa > 0$ is the mean-reversion speed of the variance process,
- $\theta > 0$ is the long-run (unconditional) variance,
- $\sigma_v > 0$ is the volatility of variance (“vol-of-vol”),

- $W_t^{(1)}, W_t^{(2)}$ are standard Brownian motions under $\mathbb{Q}$ with instantaneous correlation $\rho \in (-1, 1)$:

$$dW_t^{(1)} dW_t^{(2)} = \rho dt.$$

2.2 The Feller Condition

The variance process v_t is a Cox–Ingersoll–Ross (CIR) process. It remains strictly positive almost surely when the Feller condition holds:

$$2\kappa\theta > \sigma_v^2.$$

The calibration enforces this condition by construction via a parameter reparameterisation. This guarantees every candidate solution evaluated by the optimiser satisfies the Feller condition, so the Euler–Maruyama floor is never needed in practice.

2.3 Interpretation of Parameters

Parameter	Interpretation	Typical Range
κ	Speed at which v_t reverts to θ	(0.5, 20)
θ	Long-run average variance (annualised)	(0.01, 0.25)
σ_v	Volatility of the variance process	(0.05, 1.5)
ρ	Price-variance correlation (typically negative)	(−0.99, 0)
v_0	Initial variance at $t = 0$	(0.001, 0.5)

Table 1: Heston model parameters and their interpretations.

The negative correlation $\rho < 0$ captures the leverage effect: falling asset prices tend to coincide with rising volatility. This asymmetry produces the implied volatility skew observed in equity options.

3 Monte Carlo Simulation

3.1 Euler–Maruyama Discretisation

Continuous-time SDEs can be approximated by the Euler–Maruyama scheme. We do this by partitioning $[0, T]$ into N equal steps of size $\Delta t = T/N$. Given the state (S_t, v_t) at time t , the state at $t + \Delta t$ is:

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t) \Delta t + \sigma_v \sqrt{v_t \Delta t} Z_2, \quad (3)$$

$$S_{t+\Delta t} = S_t \exp \left[\left(r - \frac{1}{2} v_t \right) \Delta t + \sqrt{v_t \Delta t} Z_1 \right], \quad (4)$$

where $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ are standard normals with correlation ρ .

3.2 Correlated Brownian Increments via Cholesky

Two independent standard normals $\tilde{Z}_1, \tilde{Z}_2 \sim \mathcal{N}(0, 1)$ are transformed using the Cholesky decomposition of the 2×2 correlation matrix $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$:

$$\mathbf{C} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \implies \mathbf{L} = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{pmatrix}.$$

The correlated increments are then:

$$Z_1 = \tilde{Z}_1, \quad Z_2 = \rho \tilde{Z}_1 + \sqrt{1 - \rho^2} \tilde{Z}_2.$$

It is straightforward to verify $\mathbb{E}[Z_1 Z_2] = \rho$.

3.3 Log-Price Formulation

The asset price update uses the log-Euler scheme. This guarantees $S_t > 0$ for all t and improves numerical stability, especially for large Δt . The correction term $-\frac{1}{2}v_t$ arises from Itô's lemma applied to $\ln S_t$.

3.4 Simulation Parameters

The implementation uses:

- $M = 20,000$ independent Monte Carlo paths,
- $N = 252$ time steps (one per trading day),
- a fixed random seed for reproducibility.

The output is the $(M \times (N + 1))$ matrices $\{S_{t_n}^{(i)}\}$ and $\{v_{t_n}^{(i)}\}$ for paths $i = 1, \dots, M$ and time steps $n = 0, 1, \dots, N$.

4 Sentiment Analysis and Variance Adjustment

4.1 Polarity Scoring

Each news article is assigned a polarity score $p \in [-1, 1]$ using the TextBlob library, which applies a lexicon-based sentiment analysis to the concatenated title and description of each article.

The daily sentiment for calendar day d is the mean polarity of all articles published on that day:

$$s_d = \frac{1}{|\mathcal{A}_d|} \sum_{a \in \mathcal{A}_d} p_{a,d},$$

where $\mathcal{A}_d$ is the set of articles published on day d . Gaps (weekends, holidays with no articles) are forward filled.

The aggregate sentiment signal used for the variance is the time-series mean over the full lookback window of L days (default $L = 20$):

$$\bar{s} = \frac{1}{L} \sum_{d=1}^L s_d.$$

4.2 Sentiment-Adjusted Long-Run Variance

The long-run variance parameter θ is adjusted as:

$$\theta_{\text{adj}} = \theta_{\text{base}} + \lambda \bar{s},$$

where λ is a user-specified sensitivity parameter (default $\lambda = -0.02$).

Interpretation. With $\lambda < 0$:

- Positive sentiment ($\bar{s} > 0$) $\Rightarrow$ lower long-run variance.
- Negative sentiment ($\bar{s} < 0$) $\Rightarrow$ higher long-run variance.

This reflects the empirical observation that negative news is correlated with elevated uncertainty and realised volatility.

A floor is placed on θ_{adj} of 10^{-6} to ensure θ_{adj} remains strictly positive.

5 Initial Variance Estimation

5.1 Realised Variance

The initial variance v_0 is estimated from the most recent $n = 21$ trading days of log-returns. Let $r_t = \ln(S_t/S_{t-1})$. The annualised realised variance is:

$$\hat{v}_0 = 252 \times \frac{1}{n-1} \sum_{t=T-n+1}^T (r_t - \bar{r})^2,$$

where $\bar{r}$ is the sample mean over the window. A floor of 0.001 (approximately 3.2% annualised volatility) is applied.

5.2 Exponentially Weighted Moving Average (EWMA) Variance

The framework also provides an EWMA estimator:

$$\hat{\sigma}_t^2 = \lambda \hat{\sigma}_{t-1}^2 + (1 - \lambda) r_{t-1}^2,$$

with a decay factor $\lambda = 0.94$, corresponding to an effective lookback of approximately $\frac{1}{1-\lambda} = 17$ trading days. The annualised EWMA variance is $252 \times \hat{\sigma}_T^2$.

6 Model Calibration

6.1 Objective Function

Given K liquid market call prices $\{C_k^{\text{mkt}}\}_{k=1}^K$ at strikes $\{K_k\}$, the calibration minimises the mean squared error between the model and the market prices. The free parameters are (κ, f, ρ, v_0) , where f is a dimensionless Feller fraction defined below. The objective is:

$$\mathcal{L}(\kappa, f, \rho, v_0) = \frac{1}{K} \sum_{k=1}^K \left(C_k^{\text{model}}(\kappa, f, \rho, v_0) - C_k^{\text{mkt}} \right)^2,$$

where C_k^{model} is computed via Monte Carlo. Note that v_0 is a free calibration parameter rather than being fixed at the realised variance estimate. This is essential for short maturities where v_0 is the dominant driver of option prices.

6.2 Feller-Safe Reparameterisation

Instead of optimising directly over σ_v and applying a penalty if the Feller condition is violated, the optimiser searches over a Feller fraction $f \in (0, 1)$ and σ_v is recovered via the transformation:

$$\sigma_v = f \times \sqrt{2\kappa\theta}.$$

Substituting into the Feller condition:

$$\sigma_v^2 = f^2 \times 2\kappa\theta < 2\kappa\theta, \quad \text{since } f < 1.$$

The condition $2\kappa\theta > \sigma_v^2$ therefore holds for every candidate (κ, f) the optimiser evaluates, regardless of the values chosen.

The Feller margin reported after calibration quantifies how far inside the constraint the solution lies:

$$\Delta_{\text{Feller}} = 2\kappa^*\theta - (\sigma_v^*)^2 = (1 - f^{*2}) \cdot 2\kappa^*\theta > 0.$$

6.3 Strike Selection

Before calibration, market options are filtered to include only liquid strikes:

1. Retain options with a strictly positive bid: $b_k > 0$.
2. Restrict to strikes in the range $[0.6 S_0, 1.4 S_0]$ to avoid deep in-the-money or out-of-the-money options where the model is poorly identified.
3. Sort by moneyness $|K_k - S_0|/S_0$ and take the closest 15 strikes to at-the-money.

The market mid-price is defined as:

$$C_k^{\text{mkt}} = \frac{b_k + a_k}{2},$$

where b_k and a_k are the bid and ask quotes respectively.

6.4 Differential Evolution Optimiser

The loss $\mathcal{L}$ is non-convex and stochastic (Monte Carlo noise means two evaluations at the same point may return slightly different values). Therefore, the implementation uses Differential Evolution (DE), a population-based global optimisation algorithm.

Feasible bounds for the calibrated parameters are:

Parameter	Interpretation	Bounds
κ	Mean-reversion speed	(0.5, 20)
f	Feller fraction	(0.01, 0.99)
ρ	Price-variance correlation	(−0.99, 0)
v_0	Initial variance	$(0.2 \hat{v}_0, \min(3\hat{v}_0, 0.5))$

Table 2: Optimisation bounds for the differential evolution calibration.

7 Option Pricing

7.1 Risk-Neutral Monte Carlo Pricing

Under the risk-neutral measure, the price of a European call option with strike K and maturity T is:

$$C(K, T) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max(S_T - K, 0)].$$

The Monte Carlo estimate using M simulated terminal prices $\{S_T^{(i)}\}_{i=1}^M$ is:

$$\hat{C}(K, T) = e^{-rT} \frac{1}{M} \sum_{i=1}^M \max(S_T^{(i)} - K, 0).$$

Similarly for European puts:

$$\hat{P}(K, T) = e^{-rT} \frac{1}{M} \sum_{i=1}^M \max(K - S_T^{(i)}, 0).$$

7.2 Monte Carlo Standard Error

The standard error of the call price estimate is:

$$\text{SE}[\hat{C}] = \frac{e^{-rT} \hat{\sigma}_{\text{payoff}}}{\sqrt{M}},$$

where $\hat{\sigma}_{\text{payoff}}$ is the sample standard deviation of the discounted payoffs $\max(S_T^{(i)} - K, 0)$. Approximate 95% confidence intervals are reported as $\hat{C} \pm 1.96 \text{SE}[\hat{C}]$. With $M = 20,000$ paths, the standard error is typically less than 1% of the option price for near-the-money strikes.

7.3 Put-Call Parity

For any model that correctly prices derivatives under the risk-neutral measure, put-call parity must hold:

$$C(K, T) - P(K, T) = S_0 - Ke^{-rT}.$$

The framework computes the put-call parity residual:

$$\varepsilon_{\text{PCP}} = (\hat{C} - \hat{P}) - (S_0 - Ke^{-rT}).$$

A residual close to zero confirms internal consistency of the Monte Carlo estimates. Non-zero residuals arise purely from simulation noise and vanish as $M \rightarrow \infty$.

7.4 Black–Scholes Benchmark

For comparison and as a fallback when the market option chain is illiquid, the framework also computes the Black–Scholes call price:

$$C_{\text{BS}}(S_0, K, r, T, \sigma) = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2),$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

and $\Phi(\cdot)$ is the standard normal CDF. The proxy volatility $\sigma = \sqrt{v_0}$ is used.

8 Time to Expiry

The time to expiry is computed under the trading-day convention:

$$T = \max\left(\frac{N_{\text{cal}}}{252}, \frac{1}{252}\right),$$

where N_{cal} is the number of calendar days from today to the expiry date, and 252 is the assumed number of trading days per year. The floor of $1/252$ prevents degenerate zero-maturity pricing.

A separate variable $T_{\text{calib}} = \max(T, 5/252)$ is used for calibration and simulation to avoid numerical instability when the nearest expiry is fewer than five trading days away.

9 Pipeline Summary — AAPL (All Figures in USD)

9.1 Calibrated Parameters

The following results were obtained from the final calibration run on AAPL with expiry 20/03/2026 ($T = 0.0873$ years, 22 calendar days):

Parameter	Symbol	Calibrated Value
Mean-reversion speed	κ	19.995
Long-run variance	θ	0.036963
Vol-of-vol	σ_v	1.204
Correlation	ρ	-0.539
Initial variance	v_0	0.04892
Feller margin	Δ_{Feller}	0.02941
Calibration MSE	$\mathcal{L}^*$	0.01944

Table 3: Calibrated Heston parameters for AAPL (expiry 20/03/2026).

Key observations:

1. The high κ is consistent with short-dated equity options, where the market prices variance as mean-reverting almost instantaneously.
2. The positive Feller margin confirms the constraint is satisfied by construction.

9.2 Monte Carlo Simulation Results

With $M = 20,000$ paths and $N = 252$ steps over $T = 0.0873$ years, the simulated terminal price distribution gave:

$$\bar{S}_T = 275.28, \quad \hat{\sigma}_{S_T} = 16.73.$$

The risk-neutral drift check confirms correctness: $S_0 e^{rT} = 274.23 \times e^{0.048 \times 0.0873} = 275.38$, consistent with the simulated mean to within Monte Carlo noise.

9.3 Model Option Prices vs. Market

The table below shows model call and put prices for strikes near the money ($S_0 = \$274.23$), together with the put-call parity residual at $K = 275$:

Strike K	Model call $\hat{C}$	Model put $\hat{P}$	$\hat{C} - \hat{P}$
265.00	13.205	2.972	10.233
270.00	9.626	4.373	5.253
275.00	6.573	6.299	0.275
280.00	4.144	8.848	-4.704
285.00	2.395	12.079	-9.684

Table 4: Model call and put prices for AAPL near the money.

The put-call parity residual at $K = 275$ is:

$$\varepsilon_{\text{PCP}} = (6.573 - 6.299) - (274.23 - 275 e^{-0.048 \times 0.0873}) = -0.106,$$

which is negligible relative to S_0 ($\approx 0.04\%$), confirming internal Monte Carlo consistency.

9.4 Sentiment Results

Over the 20-day news lookback window, the pipeline retrieved 95 articles spanning 19 calendar days. The mean polarity was $\bar{s} = 0.1519$, indicating mildly positive news sentiment. The long-run variance adjustment was:

$$\theta_{\text{adj}} = 0.04 + (-0.02)(0.1519) = 0.036963,$$

a reduction of 0.003037 in variance units relative to base, corresponding to a decrease of approximately 0.77 percentage points in annualised implied volatility.

10 Default Parameter Values

Parameter	Symbol	Default Value
Risk-free rate	r	0.048
Mean-reversion speed	κ	2.0
Long-run variance	θ	0.04
Vol-of-vol	σ_v	0.3
Correlation	ρ	-0.7
Feller fraction (prior)	f	Free in (0.01, 0.99)
Sentiment sensitivity	λ	-0.02
Monte Carlo paths	M	20,000
Time steps	N	252
Realised var. window	n	21 days
News lookback	L	20 days

Table 5: Default parameter values used throughout the pipeline.

11 Graphical Outputs

The pipeline produces four plots, each saved as a PNG file. Upload your images to the same Overleaf project folder and replace the filenames below accordingly.

11.1 Simulated Stock Price Paths

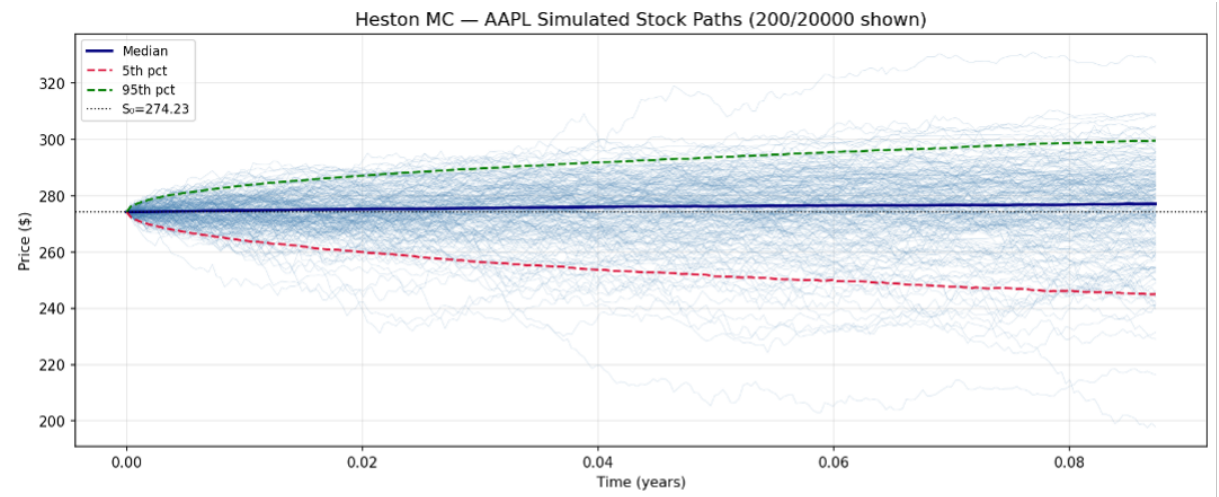


Figure 1: Heston MC — AAPL Simulated Stock Paths (200/20 000 shown). The width of the prediction band grows with t as uncertainty accumulates. Under the Heston model the distribution of terminal prices S_T is heavier-tailed than log-normal, because the variance v_t is itself random. During high-variance regimes the paths spread rapidly; inversely, during low-variance regimes they cluster tightly around the mean. The shape directly reflects the mean-reverting CIR nature of v_t .

11.2 Instantaneous Variance Paths

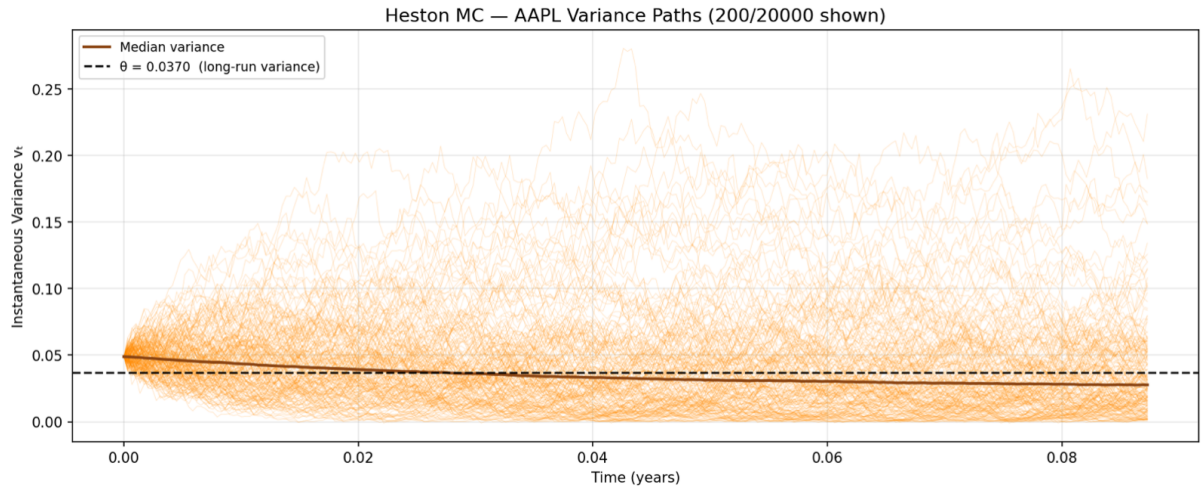


Figure 2: Heston MC — AAPL Variance Paths (200/20 000 shown). The CIR dynamics are directly visible. Paths above θ_{adj} are pulled downward by the drift $\kappa(\theta - v_t)$, while paths below are pushed upward. A large κ causes paths to cluster tightly around θ_{adj} , whereas a small κ produces slow, persistent excursions. Because the Feller condition is enforced, the variance paths remain strictly positive.

11.3 Model vs. Market Option Prices

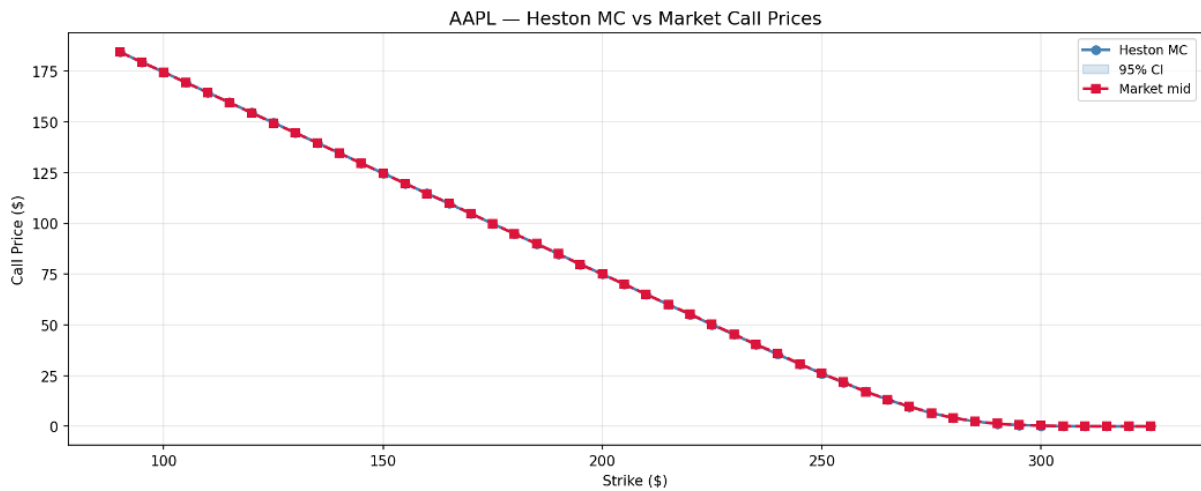


Figure 3: AAPL — Heston MC vs Market Call Prices. A well-calibrated model produces a curve that passes through or near the market mid-price markers within the confidence band. Systematic deviations indicate model misspecification or insufficient optimiser iterations. The level of alignment shown suggests that the stochastic volatility parameters are reflecting market expectations for the price distribution of AAPL with a high degree of accuracy.

11.4 Daily News Sentiment

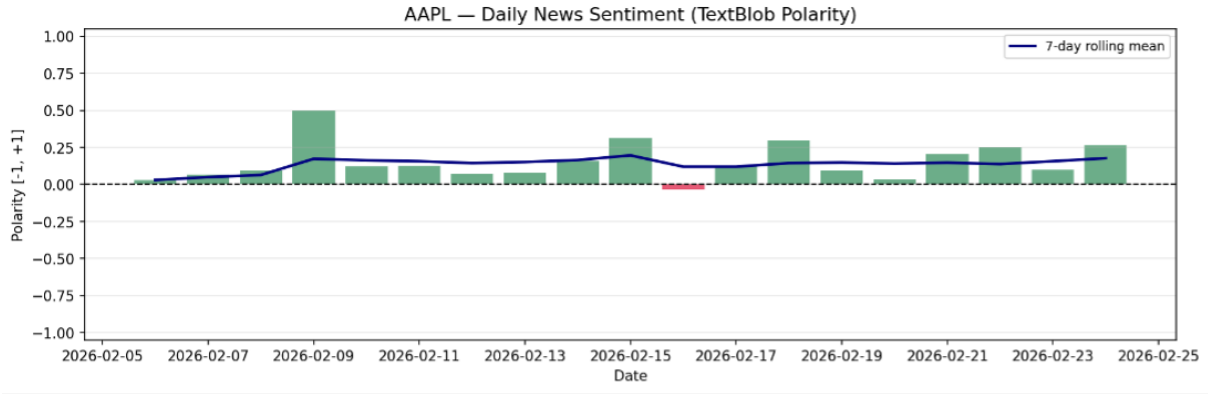


Figure 4: AAPL — Daily News Sentiment (TextBlob Polarity) over the $L = 20$ -day lookback window. Green bars denote positive polarity ($s_d > 0$) and red bars denote negative polarity ($s_d < 0$). The navy line is the 7-day rolling mean and the dashed horizontal line marks zero. The chart is predominantly green, lowering θ_{adj} below θ_{base} .

12 Limitations and Extensions

1. **Euler–Maruyama bias.** The Euler scheme introduces a discretisation error of order $\mathcal{O}(\Delta t)$. Higher-order schemes would reduce this bias at the cost of increased implementation complexity.
2. **Monte Carlo noise in calibration.** Monte Carlo noise affects calibration because the objective function is random. When the optimiser evaluates it, it is trying to minimise a loss function that contains noise rather than a smooth surface. This can make convergence slower and lead to parameter estimates that vary more.
3. **Sentiment model.** TextBlob polarity is a shallow lexicon-based measure. A transformer-based financial sentiment model (e.g. FinBERT) would likely produce more accurate scores. Summarily, the linear adjustment $\theta_{\text{adj}} = \theta + \lambda \bar{s}$ is a simplification.
4. **Single expiry.** The framework is calibrated using only one expiry date. Calibrating across several maturities instead would better restrict the parameter values and is important for pricing products whose payoffs depend on the path of the underlying asset over time.
5. **Risk-neutral vs. physical measure.** The simulation is run under the risk-neutral measure $\mathbb{Q}$. However, for risk management or P&L analysis, you would need to simulate under the real-world (physical) measure $\mathbb{P}$, which requires estimating the market price of variance risk, λ_v .

13 Conclusion

This framework provides a self-contained implementation of Heston Monte Carlo option pricing, enhanced with a news-sentiment adjustment to the long-run variance. At its core, the model is based on two linked stochastic differential equations, which are discretised using the Euler–Maruyama method with Cholesky-correlated Brownian motions. The parameters are calibrated globally to liquid market call option prices using differential evolution, and option values are calculated as the risk-neutral expected value of discounted payoffs. The sentiment component adds a forward-looking alternative data signal. It adjusts the model’s baseline volatility forecast according to the information contained in recent news.